



TB AUDIO PLUG-INS
Team Impulse Impact

USER MANUAL

TB EQ

Spectral / Dynamic Equalizer

Twenty-four bands that combine static filtering, Dynamic EQ, and spectral resonance control, so every correction happens exactly where the problem is.

24 bands

Static · Dynamic · Spectral

Stereo / Mid / Side

Windows VST3 · macOS VST3/AU



Contents

- | | | | |
|-----------|----------------------------|-----------|--------------------------|
| 01 | Overview | 07 | Listening and analysis |
| 02 | Quick start | 08 | Presets |
| 03 | Interface | 09 | Recipes |
| 04 | Working with bands | 10 | Tips and troubleshooting |
| 05 | Dynamic EQ | 11 | Parameter reference |
| 06 | Spectral resonance control | 12 | Version history |
| | | 13 | Support and license |

01 Overview

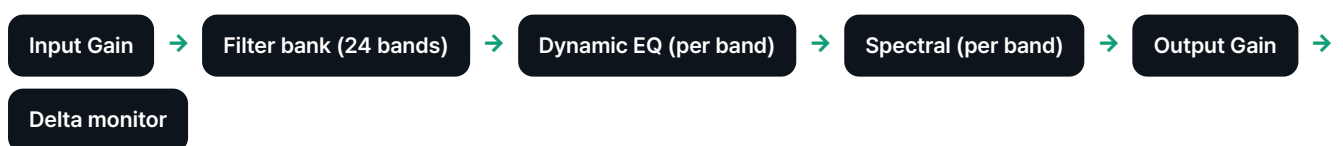
TB EQ is a 24-band equalizer that offers three kinds of processing in every band: standard filters for static tone shaping, Dynamic EQ that moves only when a frequency area gets louder than a threshold, and Spectral processing that finds and reduces narrow resonances inside the band.

Because all three live in the same band, you can fix a problem exactly where it happens. A cut that would dull a vocal all the time can become a cut that appears only on harsh syllables, and a whistling resonance can be removed without carving a permanent notch into the sound.

Key features

- 24 bands with eight filter types: Bell, Low Shelf, High Shelf, Low Cut, High Cut, Notch, Band Pass, and All Pass
- Stereo, Mid, or Side processing for each band
- Dynamic EQ on any band with **THRESHOLD**, **RANGE**, **ATTACK**, and **RELEASE**
- Spectral resonance control on any band with **SENSITIVITY**, **DEPTH**, **RANGE**, and **FOCUS**
- Delta listening that plays only what the EQ adds or removes
- Live analyzer with input and output spectra, freeze, three display ranges, and a pink-noise tilt
- Factory and User presets, with undo and redo

Signal flow



What's new in 2.0.2

- **SENSITIVITY** and **DEPTH** now work independently. SENSITIVITY decides which resonances are detected; DEPTH sets the maximum reduction.
- The spectral detail control is now labelled **FOCUS** (formerly Niddle). Parameter IDs did not change, so existing sessions, presets, and automation keep working.

About this manual

This manual describes TB EQ 2.0.2 for Windows (VST3). The current macOS build is version 1.0.3 and may not include every feature described here.

The interface image was captured from version 2.0.0. Labels that changed in 2.0.2 are described in the text.

02 Quick start

- 1 Insert TB EQ on a track. Band 1 is switched on; bands 2 to 24 start switched off.
- 2 Choose a band in the **BAND** list **3**, then pick its filter type **4** and whether it processes **Stereo**, **Mid**, or **Side** **5**.
- 3 Drag the band's node in the display **2**: left and right changes **FREQ**, up and down changes **GAIN**. Turn the mouse wheel over the node to change **Q**.
- 4 If the problem appears only when that area gets loud, switch on **Dynamic** **6** and lower **THRESHOLD** until the band reacts only to the loud moments.
- 5 If the problem is a narrow ringing or whistling tone, switch on **Spectral** instead and raise **SENSITIVITY** until the resonance is caught.
- 6 Listen to what you are changing with Delta listening (see Listening and analysis).
- 7 Match the level with **OUTPUT GAIN** **9** and compare with the bypassed signal before you judge the result.

03 Interface



TB EQ interface, captured from version 2.0.0. In 2.0.2, a band with Spectral switched on shows SENSITIVITY, DEPTH, RANGE, and FOCUS in the mode controls.

- 1 Preset bar** — previous and next preset (hold to browse), the Factory and User preset list, Save, Undo, and Redo.
- 2 EQ display** — the combined curve, band nodes, each band's dynamic range, and the live analyzer.
- 3 BAND** — chooses which of the 24 bands the controls below edit.
- 4 Filter type** — Bell, Low Shelf, High Shelf, Low Cut, High Cut, Notch, Band Pass, or All Pass.
- 5 Channel** — Stereo, Mid, or Side processing for the band.
- 6 Band switches** — **Enabled** turns the band on. **Dynamic** and **Spectral** add level-dependent or resonance processing.
- 7 Band controls** — **FREQ**, **GAIN**, and **Q**. Depending on the filter type, the second knob may read GAIN, ATTN, or SLOPE.
- 8 Mode controls** — with Dynamic on: **THRESHOLD**, **RANGE**, **ATTACK**, **RELEASE**. With Spectral on: **SENSITIVITY**, **DEPTH**, **RANGE**, **FOCUS**.
- 9 Input and output** — **INPUT GAIN** and **OUTPUT GAIN** for the whole plug-in.

04 Working with bands

Every band has the same controls, so once you know one band you know all 24. Switch a band on with **Enabled**. A band that is switched off keeps its settings but does not affect the sound.

In the display, drag a node to change frequency and gain, and turn the mouse wheel over it to change Q. Hints such as **GRAB** and **HOLD R: SOLO** appear in the display to show what the mouse will do. The curve's vertical scale can be switched between ± 24 dB and ± 48 dB for fine or large moves.

Filter types

Type	What it does	Typical use
Bell	Boosts or cuts a region around FREQ. Q sets the width.	General tone shaping and cutting resonances
Low Shelf	Raises or lowers everything below FREQ.	Warmth and low-end balance
High Shelf	Raises or lowers everything above FREQ.	Air and brightness
Low Cut	Removes content below FREQ.	Rumble, handling noise, room for the bass
High Cut	Removes content above FREQ.	Hiss, harsh top end, lo-fi effects
Notch	A very narrow, deep cut at FREQ.	Hum, whistles, a single ringing frequency
Band Pass	Keeps only the region around FREQ.	Telephone and radio effects, finding problem areas
All Pass	Shifts phase around FREQ without changing the level.	Phase alignment between related tracks

Stereo, Mid, and Side

Stereo processes both channels. **Mid** processes only what both channels share (the center), and **Side** processes only the difference between them (the width). Mid and Side bands let you brighten the edges of a mix without making the lead vocal harsher, or clean the low end of the sides while the centered bass stays intact.

Side content cancels when a mix is played in mono. After large Side changes, check the result in mono.

05 Dynamic EQ

With **Dynamic** switched on, a band changes its gain according to the level of its own frequency area. When that area rises above **THRESHOLD**, the band moves away from its **GAIN** setting by up to **RANGE**. Negative RANGE values cut, and positive values boost.

The shaded area in the display shows how far each band can move. In the interface image, GAIN is -4.2 dB and RANGE is -7.8 dB, so the band can travel from -4.2 dB down to about -12 dB, which is the dashed line.

Control	Range · default	What it does
THRESHOLD	-60 to 0 dB · -24 dB	The level at which the band starts to move. Lower values react to quieter signals.
RANGE (+BOOST / -CUT)	-24 to $+24$ dB · -6 dB	The maximum movement away from GAIN. Negative values cut, positive values boost.
ATTACK	0.1 to 200 ms · 15 ms	How quickly the band reacts. Fast settings catch peaks but can soften the start of notes.

Control	Range · default	What it does
RELEASE	1 to 2000 ms · 150 ms	How quickly the band returns. Short settings can pump; long settings can hold down the next note.
Ratio (host parameter)	1:1 to 20:1 · 2:1	How strongly the band reacts above the threshold. It is not on the panel; find it as Dynamic Ratio in your DAW's automation or parameter list.

Start with RANGE between –3 and –6 dB, then lower THRESHOLD until the band moves only on the moments you want to control.

06 Spectral resonance control

Spectral mode analyzes the band's area in short FFT frames and looks for narrow components that rise above their neighboring frequencies, such as ringing, whistles, or harsh peaks. It reduces only those components, so the rest of the band keeps its natural tone.

Control	Range · default	What it does
SENSITIVITY	0 to 1 · 0.5	How easily a resonance is detected. Higher values catch smaller peaks but can also react to tone you want to keep.
DEPTH	0 to 24 dB · 24 dB	The maximum reduction applied to a detected resonance. Since 2.0.2 it works independently of SENSITIVITY.
RANGE	0.05 to 8 · 1.0	How wide an area around FREQ is analyzed. Wide settings can also move neighboring frequencies.
FOCUS	0 to 100 · 0	Concentrates the reduction on narrower, more prominent peaks. Host automation still lists it as Spectral Niddle.

For a natural result, lower DEPTH to 6 to 12 dB first, then raise SENSITIVITY until the ringing is caught. Set Delta Mode to Selected band to hear exactly what is being removed.

Spectral processing uses more CPU than the standard filters and can behave differently on fast transients. Switch it on only for the bands that need it.

07 Listening and analysis

Delta listening

Delta listening plays only the difference TB EQ makes, which tells you whether you are removing the problem or useful tone. Delta Mode and Delta Band are host parameters: set them from an automation lane or the plug-in's parameter list in your DAW.

Parameter	Values · default	What you hear
Delta Mode	Off, All bands, Selected band · Off	Off plays the normal output. All bands plays everything TB EQ changes. Selected band plays only the change made by the band chosen in Delta Band.
Delta Band	Band 1 to 24 · Band 1	The band you hear when Delta Mode is set to Selected band.

Band audition

The display can also audition the band you are working on. Its labels show what you are hearing: **SOLO**, **BAND-PASS AUDITION** (the band's frequency area), **PHASE DELTA (AP - DRY)** for All Pass bands, and **DELTA RESONANCE** for the resonances that Spectral processing removes.

Analyzer

Control	Function
PRE	Shows or hides the input spectrum.
POST	Shows or hides the processed spectrum.
FREEZE	Freezes the analyzer display so you can study a moment.
RANGE	Cycles the analyzer range between 60, 90, and 120 dB.
SPEED	Cycles the response speed between FAST, MED, and SLOW.
TILT	RAW shows the measured spectrum. HUMAN tilts it by 4.5 dB per octave so that pink-noise-like material reads flat.
Curve scale	Switches the EQ curve's vertical scale between ± 24 dB and ± 48 dB.

08 Presets

Use the arrows in the preset bar **1** to step through presets, and hold an arrow to browse quickly. The list contains the Factory presets and your own User presets. Save stores the current settings as a User preset, a .tbeq file in the User folder. Undo and Redo step back and forward through your recent edits.

Factory presets: **Init / Flat**, **Mix Bus Polish**, **Vocal Resonance Guard**, **Master De-Harsh**, **Drum Punch & Bite**, and **M/S Clarity**.

Treat a preset as a starting point. Check each band's frequency and threshold against your own material.

09 Recipes

These settings are starting points for common jobs, not fixed answers. Adjust them by ear on your own material.

Tame harsh vocals

Harshness that appears only on loud or bright syllables.

Type	Bell
FREQ	2.5 to 5 kHz
Q	2 to 4
Mode	Dynamic
RANGE	-3 to -6 dB
ATTACK / RELEASE	5 to 15 ms / 80 to 150 ms

- 1 Find the harsh area with a temporary boost, then set GAIN back to 0 dB.
- 2 Lower THRESHOLD until the band moves only on harsh syllables.

Remove ringing resonances

Whistles, ringing drum shells, or room resonances.

Type	Bell
FREQ	At the ringing area
Mode	Spectral
DEPTH	6 to 12 dB
SENSITIVITY	Start at 0.5
FOCUS	Raise for very narrow peaks

- 1 Set Delta Mode to Selected band and raise SENSITIVITY until you hear only the ringing.
- 2 Set Delta Mode back to Off and compare with bypass.

Clean the low end

Rumble and handling noise on vocals and instruments without deep bass.

Type	Low Cut
FREQ	40 to 100 Hz
Channel	Stereo

- 1 Raise FREQ until the sound gets thin, then back off slightly.

Brighten the width, not the center

Adding air to a mix without making the lead vocal harsh.

Type	High Shelf
FREQ	8 to 12 kHz
GAIN	+1 to +2 dB
Channel	Side

- 1 Check the result in mono, where Side changes disappear.

10 Tips and troubleshooting

Symptom	What to check
It sounds better, but only because it is louder.	Match the level with OUTPUT GAIN and compare with bypass at the same loudness.
A narrow boost rings or whistles.	Lower Q or the boost. Cutting a neighboring area often sounds cleaner than boosting.
A band has no effect.	Make sure the band is Enabled. A Dynamic band also stays still until its area crosses THRESHOLD.
Width changes disappear in mono.	Side processing cancels in mono. Reduce Side changes or use Stereo instead.
CPU load is high.	Switch Spectral off on bands that do not need it.

Symptom	What to check
Ratio and Delta Mode are not on the panel.	They are host parameters. Open the plug-in's parameter list or an automation lane in your DAW.
My DAW lists FOCUS as Spectral Niddle.	That is the original parameter name, kept so existing sessions still load.

11 Parameter reference

These are the plug-in's actual ranges and defaults. Band parameters exist for each of the 24 bands and appear in your DAW with the band number.

Panel label	Host parameter	Range	Default
GLOBAL			
INPUT GAIN	Input Gain	−24 to +24 dB	0 dB
OUTPUT GAIN	Output Gain	−24 to +24 dB	0 dB
Host only	Delta Mode	Off, All bands, Selected band	Off
Host only	Delta Band	Band 1 to 24	Band 1
EACH BAND (1 TO 24)			
Enabled	Enabled	Off, On	Band 1 on, others off
Filter type	Type	Bell, Low Shelf, High Shelf, Low Cut, High Cut, Notch, Band Pass, All Pass	Bell
Channel	Channel Mode	Stereo, Mid, Side	Stereo
FREQ	Frequency	20 Hz to 20 kHz	1 kHz
GAIN	Gain	−24 to +24 dB	0 dB
ATTEN	Attenuation	−96 to −3 dB	−96 dB
Q	Q	0.1 to 30	1.0
DYNAMIC			
Dynamic	Dynamic	Off, On	Off
THRESHOLD	Dynamic Threshold	−60 to 0 dB	−24 dB
RANGE	Dynamic Range	−24 to +24 dB	−6 dB
ATTACK	Dynamic Attack	0.1 to 200 ms	15 ms
RELEASE	Dynamic Release	1 to 2000 ms	150 ms
Host only	Dynamic Ratio	1:1 to 20:1	2:1
SPECTRAL			
Spectral	Spectral	Off, On	Off
SENSITIVITY	Spectral Sensitivity	0 to 1	0.5
DEPTH	Spectral Depth	0 to 24 dB	24 dB
RANGE	Spectral Range	0.05 to 8	1.0

Panel label	Host parameter	Range	Default
FOCUS	Spectral Niddle	0 to 100	0

12 Version history

Version	Changes
2.0.2 · 2026-09-30	SENSITIVITY and DEPTH now work independently. The Niddle control is shown as FOCUS. Parameter IDs and the VST3 class ID are unchanged.
2.0.1 · 2026-08-11	The spectrum analyzer now starts as soon as the editor opens, without clicking SPEED. Version and vendor information corrected.

13 Support and license

Report problems or request features at github.com/TaebeumKim/TB_Audio-Plugins/issues. Include the plug-in version, your DAW, and your operating system.

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